

G-CAF

GovDigital Composable Architecture Framework

Volume 14

Architecture Principles Deep Dive

v2.0

Table of Contents

Chapter 1 Principle Catalog Overview

17 principles across 4 categories with theory references

Chapter 2 Core Principles (ARC-001 to ARC-010)

API-First, Cloud-Native, Security by Design, Privacy, Zero Trust, Accessibility, Open Standards, Reuse, Loose Coupling, Design for Failure

Chapter 3 Evolvability Principles (ARC-011 to ARC-013)

Interface Stability, Tolerant Evolution, Versioned Compliance — NS-aligned

Chapter 4 Operational Excellence Principles (ARC-014 to ARC-017)

Citizen Experience, Data Quality, Observable by Default, Sustainable by Design

Chapter 5 PBC Classification Deep Dive

6 PBC Types, PBC Sizing (Goldilocks), Maturity, Anti-Patterns

Chapter 6 Criticality, Sourcing & Stability Classifications

4 criticality tiers, 3 sourcing options, 3 stability tiers

Chapter 7 Artefact Lifecycle Classification

Reference Architecture, Extension, and Catalog lifecycle stages

Chapter 1

Principle Catalog Overview

All PBCs must adhere to 17 architecture principles validated at PBC Gateways. Each principle has its own module for bounded maintenance — it can evolve independently based on its underlying theory.

ID	Principle	Owner	Business Drivers	Theory
ARC-001	API-First	EA Office	BD-002, BD-008, BD-009	api-first-principle.mod
ARC-002	Cloud-Native First	EA Office	BD-001, BD-002	cloud-native-principle.mod
ARC-003	Security by Design	Security Office	BD-003, BD-004	zero-trust-principle.mod
ARC-004	Privacy by Design	Data Governance	BD-003, BD-004	—
ARC-005	Zero Trust	Security Office	BD-003	zero-trust-principle.mod
ARC-006	Accessibility First	UX Office	BD-004, BD-005	—
ARC-007	Open Standards Preferred	EA Office	BD-001, BD-008	—
ARC-008	Reuse Before Build	EA Office	BD-001, BD-002	pbcs-theory.mod
ARC-009	Loosely Coupled	EA Office	BD-002, BD-006	loose-coupling-principle.mod
ARC-010	Design for Failure	EA Office	BD-006	design-for-failure.mod
ARC-011	Interface Stability	EA Office	BD-007, BD-009	interface-stability-theory.mod
ARC-012	Tolerant Evolution	EA Office	BD-007	interface-stability-theory.mod
ARC-013	Versioned Compliance	EA Office	BD-004, BD-007	—
ARC-014	Citizen Experience First	UX Office	BD-005, BD-013	—
ARC-015	Data Quality First	Data Office	BD-010	—
ARC-016	Observable by Default	Platform Ops	BD-006, BD-011	sre-theory.mod
ARC-017	Sustainable by Design	Platform Ops	BD-011, BD-012	—

Chapter 2

Core Principles (ARC-001 to ARC-010)

ARC-001 — API-First

"All PBCs expose APIs before building UIs."

REQ-ID	Requirement	Mandatory	Validated At
REQ-INT-001	OpenAPI 3.0+ specification before implementation	Yes	Design Gateway
REQ-INT-002	API published to GovDX catalog	Yes	Go-Live Gateway
REQ-INT-003	No UI-only PBCs (APIs required)	Yes	Initiation Gateway
REQ-INT-004	API versioning strategy defined	Yes	Design Gateway

Enables composability, multi-channel support, automation, and future-proofing.

ARC-002 — Cloud-Native First

"Design for cloud; on-premise deployment only by ARB exception."

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-040	Containerized deployment (Docker/OCI)	Yes	Design Gateway
REQ-ARC-041	Kubernetes-compatible orchestration	Yes	Design Gateway
REQ-ARC-042	Stateless application tier	Yes	Design Gateway
REQ-ARC-043	External state (databases, caches)	Yes	Design Gateway
REQ-ARC-044	12-Factor App compliance	Rec	Go-Live Gateway

ARC-003 — Security by Design

"Security is embedded from inception, not bolted on later."

REQ-ID	Requirement	Mandatory	Validated At
REQ-SEC-010	Threat modeling completed	Yes	Design Gateway
REQ-SEC-011	Security controls documented	Yes	Design Gateway
REQ-SEC-012	Secure coding practices	Yes	Go-Live Gateway
REQ-SEC-013	Penetration testing	Yes	Go-Live Gateway
REQ-SEC-014	Security review sign-off	Yes	Go-Live Gateway

Fix early vs. fix in production: 10–100x cheaper.

ARC-004 — Privacy by Design

"Data protection is built into PBC architecture."

REQ-ID	Requirement	Mandatory	Validated At
REQ-DAT-010	Privacy Impact Assessment	Yes	Design Gateway
REQ-DAT-011	Data minimization documented	Yes	Design Gateway
REQ-DAT-012	Consent mechanisms implemented	Yes	Go-Live Gateway

Implements Cavoukian's 7 Foundational Principles: proactive, default, embedded, full functionality, end-to-end, visibility, respect.

ARC-005 — Zero Trust

"Never trust, always verify. Assume breach."

REQ-ID	Requirement	Mandatory	Validated At
REQ-SEC-001	MFA required for all access	Yes	Design Gateway
REQ-SEC-002	Microsegmentation implemented	Yes	Go-Live Gateway
REQ-SEC-003	Least privilege access enforced	Yes	Go-Live Gateway
REQ-SEC-004	Continuous verification enabled	Yes	Go-Live Gateway

5 pillars: Identity (UAE PASS/MFA), Device (health verification), Network (microsegmentation), Application (API gateway), Data (classification-based access).

ARC-006 — Accessibility First

"WCAG 2.1 AA compliance for all public-facing PBCs."

REQ-ID	Requirement	Mandatory	Validated At
REQ-EXP-010	WCAG 2.1 AA compliance	Yes	Go-Live Gateway
REQ-EXP-011	Accessibility testing	Yes	Go-Live Gateway
REQ-EXP-012	Keyboard navigation	Yes	Design Gateway
REQ-EXP-013	Screen reader compatibility	Yes	Design Gateway

ARC-007 — Open Standards Preferred

"Prefer open over proprietary. Avoid vendor lock-in."

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-050	Open standard preferred for new PBCs	Yes	Design Gateway
REQ-ARC-051	Proprietary choice documented with justification	Yes	Design Gateway
REQ-ARC-052	Exit strategy for proprietary technologies	Yes	Design Gateway
REQ-ARC-053	Data portability ensured	Yes	Go-Live Gateway

ARC-008 — Reuse Before Build

"Check the PBC Catalog before building new."

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-001	PBC Catalog search completed	Yes	Initiation Gateway
REQ-ARC-002	Reuse assessment documented	Yes	Initiation Gateway

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-003	Build justification (if not reusing)	Yes	Initiation Gateway
REQ-ARC-004	Shared PBC contribution (if building)	Rec	Go-Live Gateway

ARC-009 — Loosely Coupled

"Minimize dependencies. Enable independent deployment."

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-010	No shared database across PBCs	Yes	Design Gateway
REQ-ARC-011	API-based communication only	Yes	Design Gateway
REQ-ARC-012	Async preferred for non-blocking operations	Rec	Design Gateway
REQ-ARC-013	Circuit breakers for dependencies	Yes	Go-Live Gateway
REQ-ARC-014	Dependency count documented	Yes	Go-Live Gateway

ARC-010 — Design for Failure

"Assume PBCs will fail. Design for graceful degradation."

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-060	Failure modes documented	Yes	Design Gateway
REQ-ARC-061	Circuit breakers implemented	Yes	Go-Live Gateway
REQ-ARC-062	Graceful degradation defined	Yes	Go-Live Gateway
REQ-ARC-063	Health checks configured	Yes	Go-Live Gateway
REQ-ARC-064	Recovery procedures documented	Yes	Go-Live Gateway

Chapter 3

Evolvability Principles (ARC-011 to ARC-013)

These three principles implement Normalized Systems theorems, ensuring bounded changes produce bounded work.

ARC-011 — Interface Stability

"Separate stable contracts from internal evolution."

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-020	Interface contract published separately	Yes	Design Gateway
REQ-ARC-021	Internal changes don't break consumers	Yes	Go-Live Gateway
REQ-ARC-022	Semantic versioning for interfaces	Yes	Design Gateway
REQ-ARC-023	Deprecation windows enforced	Yes	Go-Live Gateway

Implements NS Action Version Transparency.

ARC-012 — Tolerant Evolution

"Accept unknown fields; ignore what you don't need."

REQ-ID	Requirement	Mandatory	Validated At
REQ-ARC-030	Tolerant reader implemented	Yes	Design Gateway
REQ-ARC-031	Unknown fields ignored (not rejected)	Yes	Design Gateway
REQ-ARC-032	Schema validation uses open model	Yes	Go-Live Gateway

Implements NS Data Version Transparency. Enables non-breaking schema evolution.

ARC-013 — Versioned Compliance

"Declare specific standard contract versions."

REQ-ID	Requirement	Mandatory	Validated At
REQ-INT-040	Contract versions declared at Initiation	Yes	Initiation Gateway
REQ-INT-041	Version compliance tracked in Catalog	Yes	Go-Live Gateway
REQ-INT-042	Migration plan for version upgrades	Yes	Go-Live Gateway

Implements NS Action Version Transparency. PBCs don't break from surprise updates.

Chapter 4

Operational Excellence (ARC-014 to ARC-017)

Added in v1.18 to address citizen experience, data quality, observability, and sustainability.

ARC-014 — Citizen Experience First

"Design services for citizens, not systems."

REQ-ID	Requirement	Mandatory	Validated At
REQ-EXP-001	User research conducted	Yes	Initiation Gateway
REQ-EXP-002	UX review for citizen-facing PBCs	Yes	Design Gateway
REQ-EXP-003	Task completion rate measured	Yes	Go-Live Gateway
REQ-EXP-004	Feedback mechanism implemented	Yes	Go-Live Gateway
REQ-EXP-005	Non-digital alternative documented	Yes	Go-Live Gateway

ARC-015 — Data Quality First

"Data is only valuable if it's trusted."

REQ-ID	Requirement	Mandatory	Validated At
REQ-DAT-001	Data quality dimensions defined	Yes	Design Gateway
REQ-DAT-002	Source of Truth (SoT) assignment	Yes	Design Gateway
REQ-DAT-003	Quality metrics configured	Yes	Go-Live Gateway
REQ-DAT-004	Data lineage documented	Yes	Go-Live Gateway
REQ-DAT-005	Quality SLA with consumers	Rec	Go-Live Gateway

Critical for AI-readiness: Agent and Analytics PBCs require quality data.

ARC-016 — Observable by Default

"If you can't see it, you can't fix it."

REQ-ID	Requirement	Mandatory	Validated At
REQ-OPS-001	Structured logging implemented	Yes	Design Gateway
REQ-OPS-002	Metrics endpoint exposed	Yes	Go-Live Gateway
REQ-OPS-003	Distributed tracing enabled	Yes	Go-Live Gateway
REQ-OPS-004	Health endpoints available	Yes	Go-Live Gateway

REQ-ID	Requirement	Mandatory	Validated At
REQ-OPS-005	Alerting configured	Yes	Go-Live Gateway

ARC-017 — Sustainable by Design

"Build efficient systems. Minimize resource consumption."

REQ-ID	Requirement	Mandatory	Validated At
REQ-OPS-010	Resource sizing justified	Yes	Design Gateway
REQ-OPS-011	Auto-scaling configured	Rec	Go-Live Gateway
REQ-OPS-012	Utilization monitored	Yes	Go-Live Gateway
REQ-OPS-013	Efficiency review (Tier 1 PBCs)	Yes	Annually

Chapter 5

PBC Classification Deep Dive

6 PBC Types

Type	Primary Function	Data Relationship	Examples
Application	Business object lifecycle (CRUD)	Owns entity data; Source of Truth	Citizen, License, Case, Payment
Analytics	Insight, predictions, AI services	Consumes from Data Fabric; no writes	Overdraft, Fraud Detection
Data	Reference data, consolidated queries	Operates on Data Fabric; delivers data	Address Validation, Knowledge Graph
Orchestration	Multi-PBC process coordination	Manages process state only	Application Processing, Multi-Agency
Digital Twin	Digital representation of physical assets	Owns asset model and time-series data	Smart Meter, Traffic Sensor, Fleet
Agent	Autonomous AI with reasoning	No entity ownership; operates via contracts	Customer Service, Document Processing

PBC Sizing — The Goldilocks Principle

A PBC must be large enough to deliver standalone business value, and small enough to remain manageable and replaceable. Too small: no standalone value, excessive inter-PBC communication, micro-service scale. Too large: multiple teams needed, different internal change rates, technology-led definition.

PBC Anti-Patterns

Anti-Pattern	Problem	Solution
Excessive Decomposition	Integration complexity; chattiness	Merge related capabilities; apply sizing
Little Monoliths	PBC too coarse; hard to replace parts	Split by change frequency; team alignment
Technology-Led Definition	Organized by tech stack, not capability	Restart from business capabilities
Org-Chart Mapping	Boundaries mirror org structure	Design for capabilities; let teams align
Shared Database	Tight coupling via shared data	Each PBC owns its data
Distributed Monolith	All PBCs must deploy together	Apply loose coupling; async communication

Chapter 6

Criticality, Sourcing & Stability

Criticality Levels

Level	Governance	SLA	Examples
Critical	ARB oversight; highest scrutiny	99.99%	Digital ID, Citizen Registry, Digital Payments
High	DDA priority review	99.9%	Finance, HR, Procurement platforms
Medium	Standard DDA review	99.5%	Case Management, Email, Collaboration
Low	Streamlined review	99%	Legacy Integration, Custom Analytics

Sourcing Options

Preference	Decision Authority	Typical Timeline	Key Consideration
Build	ARB (Critical) / DDA (others)	12–18 months	Full control; highest cost; custom for unique needs
Buy	DDA Technical Committee	3–6 months	Fast; vendor lock-in risk; commodity capabilities
Outsource	DDA + Procurement	6–12 months	Specialized skills; IP negotiation; knowledge transfer

Stability Tiers

Tier	Change Process	Deprecation Window	Examples
Foundational	ARB approval for breaking changes	12 months	Digital ID, Payments, Base Registries
Stable	DDA approval; consumer notification	6 months	Case Management, Notifications, Finance
Evolving	DDA notification; auto compatibility check	3 months	Experimental, Entity-specific PBCs

Chapter 7

Artefact Lifecycle Classification

Parallel to PBC maturity, framework artefacts (RAs, extensions, catalogs) have their own lifecycle stages so adopters know which are production-proven.

Stage	RA Lifecycle	Extension Lifecycle	Catalog Lifecycle
Incubating	Actively developing; expect changes	Under development; API may change	Schema evolving; limited data
Pilot	Being validated with pilot entities	Tested in limited scope	Populated with pilot data
Stable	Production-proven; recommended	Released; backward-compatible	Full data; quality-governed
Mature	Widely adopted; minimal changes	Widely adopted; feature-complete	Comprehensive; cross-linked
Deprecated	Superseded; migration path available	Replaced; migration available	Sunset planned; export available
Retired	No longer maintained	Removed	Archived

An entity deciding whether to adopt REF-003 (IDP) needs to know it's Incubating — expect changes. REF-001 (OneHub) is Stable — production-proven, recommended for adoption.

PBC Maturity (Gartner Model)

Maturity Type	Data Ownership	Autonomy	Primary Use Case
Fully Expressed	Complete	High	Core business entities; greenfield
Orchestration	Process state only	Medium	Workflow coordination
Basic Business Function	Partial	Low	Legacy modernization; incremental extraction
Pseudo PBC	None (facade)	None	Transitional; legacy wrapping

PBC Maturity is orthogonal to PBC Type. Maturity describes HOW COMPLETE a PBC is; Type describes WHAT FUNCTION it performs.